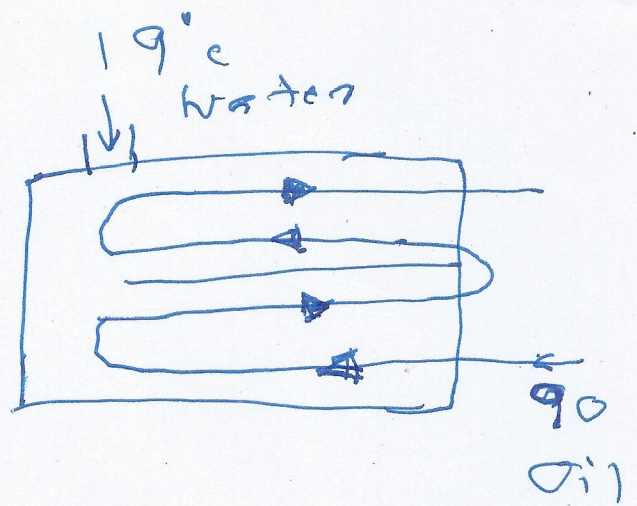


Qn. 5

$$Area = 26 m^2$$

$$C_h = 1.5 \times 2000 \\ = 3000 W/K.$$



$$C_c = 1 \times 4170 W/K.$$

$\therefore C_h$ is C_{min} , and C_c is C_{max}

$$U = 250 W/m^2 K$$

$$NTU = \frac{UA}{C_{min}} = \frac{250 \times 26}{3000} \\ = 2.167$$

$$C_{min}/C_{max} = 0.72$$

From graph for 2 shell pass & Tube

pass $\epsilon = 0.72$

now
$$\epsilon = \frac{C_c (T_{co} - T_{ci})}{C_{min} (T_{hi} - T_{ci})}$$

$$0.72 = \frac{4170 (T_{co} - 19)}{3000 (90 - 19)}$$

2)

$$T_{co} = 55.78^{\circ}\text{C}$$

Also

$$\epsilon = \frac{C_h (T_{hi} - T_{ho})}{C_{min} (T_{hi} - T_{ci})}$$

$$0.72 = \frac{3000 (90 - T_{ho})}{3000 (90 - 19)}$$

$$\therefore T_{ho} = 38.88^{\circ}\text{C}$$

$$Q = 3000 (90 - 38.88)$$

$$= \underline{153.36 \text{ kW}}$$

Off Design performance

Oil flow reduced to 75%

$$\therefore m_h C_{ph} = 0.75 \times 1.5 \times 2000$$

$$= 2250 \text{ W/K}$$

$$C_{min} \text{ is } C_h = 2250 \text{ W/K}$$

$$NTU = \frac{250 \times 26}{2250} = 2.88$$

$$C = \frac{2250}{4170} = 0.54$$

③

From graph

$$\text{Effectiveness } \epsilon = 0.8$$

$$0.8 = \frac{2250(90 - T_{ho})}{2250(90 - 19)}$$

$$\therefore T_{ho} = 33.2^\circ\text{C}$$

$$0.8 = \frac{4170(T_{co} - 19)}{2250(90 - 19)}$$

$$T_{co} = 49.65^\circ\text{C}$$

$$Q = 4170(49.65 - 19)$$

$$= \underline{\underline{127.8 \text{ kW}}}$$